



SIMATS ENGINEERING

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Course Code: CSA0208

Slot: D

Course Name: C Programming

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Project Title:

MEMORY MANAGEMENT SIMULATOR USING C

Module Photographs:

MODULE 1 — INPUT & MEMORY DATA PARSING

- ✓ The main() function receives block and process data through argv[].
- ✓ strtok() separates comma-delimited values into individual tokens.
- ✓ atoi() converts each token from text into an integer.
- ✓ Blocks are stored in blocks[50]; processes are stored in processes[50].
- ✓ The counters m and n track the number of blocks and processes.

```
char *token = strtok(argv[1], ",");
while (token != NULL) {
    blocks[m++] = atoi(token);
    token = strtok(NULL, ",");
}
```

Memory Management Simulator

Enter Block Sizes (comma separated)

Enter Process Sizes (comma separated)

Run Simulation

Output

```
Memory Management Simulator
Process No.  Process Size  Block No.
1             100         1
2             200         2
3             300         3
4             400         4
5             500         5
6             600         6
7             700         7
8             800         8
9             900         9
10            1000        10
```

Project Description:

This module is responsible for receiving and processing the input data required for the memory management system. The main() function accepts the input values through the argv[] array, allowing memory block sizes and process sizes to be provided in a structured format. The strtok() function is used to separate comma-delimited input values into individual tokens. Each token is then converted from a string into an integer using the atoi() function. The memory block sizes are stored in the blocks[50] array, while the process sizes are stored in the processes[50] array. The variables m and n are used as counters to track the total number of memory blocks and processes available. This module ensures that all input data is properly parsed and stored before memory allocation algorithms such as First Fit, Best Fit, or Worst Fit are executed.

Student Signature

Guide Signature